

ScholAR: Page-Grounded Retrieval-Augmented Generation for Scientific Document Comprehension

Anonymous submission

Abstract

Understanding and querying scientific papers demands precise, evidence-grounded comprehension that general-purpose language models struggle to provide reliably. Existing PDF-chat systems hallucinate citations, discard document structure, and cannot trace generated claims back to verifiable source pages. We present **ScholAR** (Smart Companion for **H**olistic **O**rganization, **L**iterature **A**nalysis & **R**esearch), a retrieval-augmented generation framework designed specifically for scientific document comprehension that runs **entirely on-device**, with no cloud API in the loop for either text or vision inference. ScholAR enforces strict page-boundary-preserving text segmentation, ensuring that every retrieved chunk maps exactly to a single PDF page. An indirect citation grounding mechanism decouples LLM generation from page-number prediction, replacing fragile in-context page references with verifiable evidence identifiers that the frontend resolves to precise PDF highlights. Beyond single-document text QA, ScholAR extends to **visual grounding** (routing figure/table questions to a local, natively multimodal model) and **multi-document citation-chasing** (resolving and ingesting a paper’s own references so questions can be answered against both the anchor paper and its cited works). We introduce a manually annotated retrieval benchmark spanning 14 query–chunk pairs across 3 landmark NLP papers, evaluated using Recall@ K and Mean Reciprocal Rank (MRR). Our BM25-primary retrieval system achieves Recall@5 of 0.929 and MRR of 0.788, outperforming keyword-overlap baselines by 10.1 percentage points in MRR. An NLI-based faithfulness metric (Combined Faithfulness Score, CFS) evaluated on a 51-case oracle-claim benchmark shows hybrid BM25+Dense+RRF retrieval reaching Combined CFS of 0.829 and 96.1% of claims judged faithful. On an 18-case visual-grounding pilot benchmark spanning four reasoning types, retrieval correctly routes to the target figure or table in 18/18 cases. On a 10-case multi-document locality benchmark, ScholAR retrieves the correct cited paper within the top 5 results half of the time (Recall@5 0.50, MRR 0.183), below a random-guessing floor computed over the same candidate pool (Recall@5 0.625) — a clear open challenge, consistent with prior findings that even large foundation models struggle at cross-paper localization.

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1 Introduction

Reading unfamiliar scientific papers is slow precisely where it matters most: verifying that a claimed number or method detail says what a reader (or an assistant) thinks it says. General-purpose LLM chat over an uploaded PDF makes this worse before it helps — it summarizes fluently but readily invents page numbers, since nothing constrains a citation to text the model actually saw. Existing scientific-RAG systems close part of this gap but assume resources most readers lack locally: cloud backends, multi-million-paper datastores, or both (Section 2.2). We instead ask what grounded, citable QA looks like under a hard local-only constraint: a single sub-15B model, no cloud API. ScholAR combines page-preserving chunking (every chunk maps to one PDF page) with an indirect citation mechanism (the model cites evidence IDs it actually retrieved, not free-form page numbers), extended to figures/tables and a paper’s own cited references. Concretely, we contribute:

- An end-to-end, **fully local** RAG framework for scientific document comprehension enforcing strict page-boundary-preserving text segmentation, with text and vision inference both on-device.
- An **indirect citation grounding** mechanism: the model cites only evidence IDs mapped to chunks it actually retrieved, preventing page-number hallucination by construction.
- Extensions to **visual question answering** over figures/tables and **multi-document citation-chasing**, resolving and ingesting a paper’s own references for cross-paper questions.
- Quantitative benchmarks for all four capabilities (retrieval, faithfulness, visual grounding, multi-document locality), each with manually annotated cases and Recall@ K /MRR-style metrics.
- Controlled **ablation experiments** comparing keyword overlap, BM25-only, and BM25-primary-with-reranking retrieval architectures, and comparing two local models (qwen3.5:9b, gemma4:12b) on visual grounding to characterize model choice for a fully on-device system.

2 Related Work

2.1 Retrieval-Augmented Generation

The standard RAG framework was introduced by Lewis et al. (2020). BM25 probabilistic retrieval was formalised by Robertson and Zaragoza (2009). The BEIR benchmark (Thakur et al. 2021) provides a heterogeneous zero-shot evaluation suite for retrieval systems.

2.2 Scientific Literature RAG Systems

PaperQA (Lála et al. 2023) was an early high-accuracy RAG agent for scientific QA, relying on proprietary cloud LLMs throughout; its successor PaperQA2 (Skarlinski et al. 2024) reports superhuman literature-synthesis accuracy via an agentic, multi-backend (GPT-4-/Claude-class) design. OpenScholar (Asai et al. 2024) substantially outperforms PaperQA2 and GPT-4o using an open retrieval pipeline, though generation still runs through large hosted or self-hosted models. SciRAG (Ding et al. 2025) further improves over both via citation-graph-aware symbolic reasoning. None targets a fully on-device deployment: all call a cloud API or assume large self-hosted models. ScholAR instead targets consumer hardware, running both text and vision through a single sub-15B-parameter local model, trading some capability for zero data exposure and API dependency.

2.3 Scientific Document Understanding

GROBID (Lopez 2009) and Nougat (Blecher et al. 2023) provide structured parsing of academic PDFs into machine-readable formats. SlideTailor (Zeng et al. 2026) generates personalised slides from papers but relies on the same raw PDF text extraction bottleneck ScholAR addresses. The S2ORC corpus (Lo et al. 2020) provides a large-scale resource of parsed academic papers.

2.4 Visual Document Retrieval

ColPali (Faysse et al. 2024) and its successors index each document page as a visual embedding, retrieving pages directly from image content without OCR. ScholAR takes an architecturally different, cheaper bet: BM25 text retrieval stays the primary signal for the common text-QA path, and only figure/table-targeted queries route to a vision-capable model applied to a cropped region around the identified figure.

2.5 Multi-Modal and Multi-Document Scientific QA

M3SciQA (Li et al. 2024) is the closest existing benchmark to ScholAR’s multi-document extension: it pairs each anchor paper with its cited references, requiring multi-modal and cross-document reasoning, and finds 18 foundation models all substantially underperform humans at this task. ScholAR’s smaller multi-document locality benchmark (Section 5.4) echoes this: cross-paper localization is hard even when the correct reference is already in the retrieval pool.

2.6 Question Answering over Documents

QASPER (Dasigi et al. 2021) is the closest existing benchmark to our task, covering information-seeking QA over scientific papers. SciFact (Wadden et al. 2020) evaluates fact verification over biomedical literature.

2.7 Citation Grounding and Faithfulness

Gao et al. (2023) study attribution in LLM generation; their analysis shows that citation quality degrades sharply without retrieval grounding. Min et al. (2023) propose FActScore, which decomposes generated text into atomic claims and verifies each claim against a knowledge source, adopting this decomposition strategy in our benchmark construction. Laban et al. (2022) introduce SummaC for consistency detection in summarization, with two variants: SummaC-Conv (fine-tuned NLI) and SummaC-ZS (zero-shot via sentence embeddings). We implement SummaC-ZS because it does not require a fine-tuned NLI model (e.g., DeBERTa-MNLI), making it reproducible without GPU infrastructure or API access. Zha et al. (2023) show that embedding-based alignment metrics substantially outperform ROUGE-family metrics for hallucination detection; this motivates our choice of SummaC-ZS over lexical alternatives. RAGAS (Es et al. 2024) is the closest existing automated RAG-faithfulness framework to our own CFS metric, but scores an answer against retrieved context directly rather than decomposing it into atomic claims verified against oracle evidence, which is the design choice we adopt from FActScore.

3 Problem Formulation

Given a scientific PDF D , a user question q , and a corpus of extracted chunks $\mathcal{C} = \{c_1, \dots, c_n\}$ derived from D (and, in multi-document mode, from D ’s cited references), the system must retrieve a ranked subset $\mathcal{C}_q \subseteq \mathcal{C}$ and generate an answer $\hat{a}$ grounded only in that evidence:

$$\hat{a} = \arg \max_a P(a \mid q, \mathcal{C}_q) \quad \text{where} \quad \mathcal{C}_q = \text{Retrieve}(q, \mathcal{C}, k) \quad (1)$$

Each chunk c_i carries a source page number and, in multi-document mode, a source paper identifier, so $\mathcal{C}_q$ always resolves back to a specific page of a specific document. Generation attaches each claim in $\hat{a}$ to an evidence identifier e_j referencing a specific $c_i \in \mathcal{C}_q$ (Section 4.8) rather than a free-form page number, so citations are constrained to what was actually retrieved regardless of Retrieve’s own precision. Visual grounding (Section 4.5) substitutes an image-conditioned generation step when q names a figure or table; multi-document mode (Section 4.6) enlarges $\mathcal{C}$ across papers before Retrieve is applied, without changing this formulation.

4 Method

4.1 Document Processing Pipeline

Given a paper (an arXiv identifier or an uploaded PDF), ScholAR downloads or accepts the file, extracts per-page text via PyMuPDF (Artifex Software 2024), applies page-preserving chunking (below) and figure/table extraction

Algorithm 1 Page-Preserving Chunking

Require: Pages $P = \{p_1, \dots, p_n\}$, target words W , overlap δ
Ensure: Chunks $\mathcal{C}$

```
1:  $\mathcal{C} \leftarrow []$ 
2: for each page  $p_i \in P$  do
3:    $w \leftarrow \text{split}(p_i.\text{text})$ 
4:    $s \leftarrow 0$ 
5:   while  $s < |w|$  do
6:      $e \leftarrow \min(s + W, |w|)$ 
7:      $\mathcal{C}.\text{append}(\text{Chunk}(w[s:e], p_i.\text{page}))$ 
8:      $s \leftarrow e - \delta$ 
9:   end while
10: end for
11: return  $\mathcal{C}$ 
```

(Section 4.5), and persists the results as `metadata.json`, `pages.json`, `chunks.json`, and `figures.json`: raw PDF in, page-preserving chunks with source page numbers and (in multi-document mode) source paper identifiers out, ready for retrieval and citation-grounded generation.

4.2 Page-Preserving Chunking

Standard character- or token-window chunking routinely splits a chunk across a page boundary, breaking the one-to-one mapping between a retrieved chunk and the PDF page a citation should highlight. ScholAR instead chunks strictly within each page: Algorithm 1 slides a window of $W=1400$ target words with $\delta=120$ -word overlap over each page’s token stream independently, so a chunk never spans two pages. A lightweight regex pass additionally tags each chunk with a detected section title (Abstract, Method, Results, ...) when a heading-like line is found near the top of the page; the retriever uses this as a section-hint signal below.

4.3 BM25-Primary Retrieval with Heuristic Reranking

The production retriever ranks chunks with Okapi BM25 (Robertson and Zaragoza 2009) ($k_1=1.4$, $b=0.72$) as the primary signal, then applies small additive heuristic boosts: a query-expansion dictionary maps research-specific terms to related vocabulary (e.g., *result* $\rightarrow$ *accuracy*, *BLEU*, *score*); an explicit page number adds a page-hint boost; query language implying a paper section (e.g., *limitation*, *results*) adds a section-hint boost on a matching chunk; and rhetorical phrase cues (*we propose*, *we introduce*, *outperforms*) add a small boost, since claim-bearing sentences more often satisfy an information-seeking query. Visual-cue boosting for figure/table chunks (Section 4.5) uses the same mechanism. We initially explored a design where a dense/semantic score dominated ranking; our own ablation (Table 5) showed BM25-primary more reliable on this benchmark, so semantic, section, and phrase signals stay light rerankers, while dense retrieval is used only in the separate Hybrid BM25+Dense+RRF configuration evaluated for faithfulness (Section 5.2).

4.4 Indirect Citation Grounding

Free-form citation prompting — asking the model to write, e.g., “[p. 7]” directly — is fragile: the model has no reliable way to know which page a claim came from and will readily hallucinate a plausible-looking one. ScholAR instead assigns each piece of retrieved evidence a short-lived evidence identifier ($E_1, E_2, \dots$) for the current turn, and the generation prompt instructs the model to cite *only* these IDs, never a page number directly. After generation, ScholAR discards any page number the model wrote on its own and rewrites each valid evidence-ID citation into a compact numbered reference ($[1], [2], \dots$); the frontend resolves each reference back to its source chunk’s page and highlights the corresponding PDF region. The model only decides *which* evidence supports a claim; the mapping to actual page and highlight is deterministic application logic, not a model output.

4.5 Visual Grounding

When a query is judged to reference visual content (via cue terms such as “figure,” “table,” or “plot,” or an explicit figure/table number), retrieval boosts figure and table chunks so they can compete against longer text chunks in the BM25 ranking. Each figure/table caption on a page is detected via regex, and the region above it is rendered at $3\times$ zoom and cached alongside the caption text. A query naming an explicit figure or table number (e.g., “explain figure 2”) is matched directly against that figure’s own label rather than relying on lexical overlap alone: our tokenizer, like most BM25 implementations, only begins a token on a letter, so “Figure 2” and “Figure 14” both reduce to `figure` once digits are dropped, making them otherwise indistinguishable to the ranker. When the top-ranked chunk is a figure or table, ScholAR base64-encodes the cached image and sends it, with the caption and supporting text, to the same local Ollama model used for text generation (both models are natively multimodal) rather than a separate vision model or cloud API. If the image is missing or the model call fails, the system degrades to a caption-only extractive answer rather than failing outright.

4.6 Multi-Document Extension

ScholAR resolves an anchor paper’s own bibliography and lets the user ingest individual cited works into the same study session: for arXiv papers via the Semantic Scholar API by arXiv ID; for uploaded PDFs, via an S2 title search over the last several pages, regex-matched against numbered ([N]) and author-year (Vaswani et al., 2017) citation styles. Ingesting a reference downloads its PDF, applies the same page-preserving chunking as the anchor paper, and tags every chunk with a `source_paper_id`. At query time, chunks from all loaded secondary papers merge with the anchor’s into one retrieval pool, and each citation is labeled with its source paper.

4.7 Study Goal Generation (Product Feature, Not Independently Benchmarked)

ScholAR generates eight paper-specific study goals per document using a local Ollama model (`qwen3.5:9b` or

gemma4:12b). Each goal has a title, a 2–4 sentence description grounded in paper text, and 3–5 recursive subquestions whose evidence chunks are resolved by a second BM25 retrieval pass. When no LLM is available, a fully extractive fallback generates goals directly from sentence patterns in the abstract and introduction. We describe this feature for completeness, but — unlike retrieval, faithfulness, visual grounding, and multi-document locality — we do not report a dedicated benchmark for it in this paper; it is not one of this paper’s evaluated contributions.

4.8 Faithfulness Evaluation: SummaC-ZS + Semantic Coverage

We evaluate citation faithfulness using the SummaC-ZS protocol (Laban et al. 2022), the zero-shot variant of SummaC that does not require a fine-tuned NLI model (Section 2.7 motivates this choice over ROUGE-family metrics).

Tier 1: SummaC-ZS (Semantic Entailment). Each ground-truth claim c is first decomposed into atomic subclaims $\{a_1, \dots, a_m\}$ via sentence- and clause-level splitting (FactScore-style (Min et al. 2023)). For each atom a_i and each retrieved chunk C_j , we compute:

$$s(a_i, C_j) = \max_{t \in \text{sents}(C_j)} \cos(\phi(a_i), \phi(t)) \quad (2)$$

where $\phi(\cdot)$ is the all-MiniLM-L6-v2 sentence encoder (Reimers and Gurevych 2019) loaded from the local HuggingFace cache (no network access required), and the max is over all sentences in the chunk. Atom a_i is classified ENTAILMENT if $\max_j s(a_i, C_j) \geq \tau_e$ (threshold $\tau_e = 0.42$), NEUTRAL if $\geq \tau_n = 0.25$, and CONTRADICTION otherwise. Numeric claims containing values not present in any retrieved chunk are down-graded from ENTAILMENT to NEUTRAL. τ_e , τ_n , and the Faithful cutoff (CFS ≥ 0.55 , below) were calibrated manually against a small pilot set of claims with obvious gold labels; we have not run a systematic sensitivity analysis over these thresholds, which we flag as necessary follow-up rather than treat as settled.

$$\text{SummaC-ZS}(c, C) = \frac{|\{a_i : \text{label}(a_i) = E\}|}{m} \quad (3)$$

Tier 2: SCR (Semantic Coverage Rate). Whole-claim vs. best-chunk cosine similarity (a coarser, complementary signal): $\text{SCR}(c, C) = \max_j \cos(\phi(c), \phi(C_j))$.

Tier 3: KFP (Key Fact Presence). Fraction of numeric values and technical proper nouns from the claim found verbatim in the top-5 retrieved chunks (exact lexical recall for quantities and named entities, where semantics cannot substitute).

The three tiers combine into a single Combined Faithfulness Score (CFS):

$$\text{CFS} = 0.50 \times \text{SummaC-ZS} + 0.30 \times \text{SCR} + 0.20 \times \text{KFP} \quad (4)$$

4.9 System Architecture

Scholar is built as a fully local system. The backend is a FastAPI application that stores parsed PDFs as JSON files on disk (metadata.json, pages.json, chunks.json).

The frontend is a Next.js single-page application with an embedded pdf.js viewer that highlights cited pages in real time. All inference, both text generation and figure/table visual question answering, is performed locally via Ollama using a single natively multimodal model (qwen3.5:9b or gemma4:12b); no cloud API is used anywhere in the pipeline. No user data ever leaves the local machine. All experiments in this paper were run on a consumer laptop (Apple Silicon, 18 GB unified memory) under Python 3.12, PyTorch, and Ollama, with all-MiniLM-L6-v2 (Reimers and Gurevych 2019) loaded from a local Hugging Face cache for the faithfulness pipeline; no discrete or cloud GPU was used for any reported result — inference runs via Ollama’s Metal-accelerated backend on the laptop’s own integrated GPU.

5 Evaluation

5.1 Retrieval Benchmark

We evaluate retrieval quality on a manually annotated benchmark of 14 cases spanning three landmark NLP papers: *Attention Is All You Need* (1706.03762 (Vaswani et al. 2017)), *Retrieval-Augmented Generation* (2005.11401 (Lewis et al. 2020)), and *LLaMA* (2302.13971 (Touvron et al. 2023)). Each case specifies a user-style question, a set of expected relevant chunk IDs, and the expected source pages. Query types include main idea, architecture, training details, result tables, human evaluation, safety analysis, and page-hint queries. We report Recall@K ($K \in \{1, 3, 5\}$) and Mean Reciprocal Rank (MRR).

5.2 Faithfulness Benchmark

For the faithfulness evaluation, we construct **51 oracle-claim cases** manually annotated across three landmark NLP papers: *Attention Is All You Need* (Vaswani et al. 2017) (15 cases), *RAG* (Lewis et al. 2020) (12 cases), and *LLaMA* (Touvron et al. 2023) (24 cases). Each case contains a natural-language query, a ground-truth claim, a manually verified supporting chunk ID, and a claim type label. Claim types include: result_number (13), training_detail (8), technical_claim (11), architecture_detail (10), formula (1), conceptual_claim (5), human_eval (2), and environmental_claim (1). All annotations were produced manually by the authors.

5.3 Visual Grounding Benchmark

We construct an 18-case pilot benchmark spanning four reasoning types: visual_understanding (6), data_extraction (5), comparison (4), and location (3), each specifying a figure/table-targeted question and the expected label(s). We report two signals: *retrieval grounding* — whether the correct figure or table ranks first, isolating retrieval (Section 4.5) from the vision model’s reading accuracy — and a lightweight *answer-quality* proxy (fraction of non-trivial question content words that reappear in the answer). Under gemma4:12b, retrieval grounding is 18/18 with mean answer score 0.731 and no caption-only

Table 1: Visual grounding retrieval results (18 cases). All cases correctly route to the target figure or table.

Reasoning Type	N	Retrieval R@5
Visual understanding	6	1.000
Data extraction	5	1.000
Comparison	4	1.000
Location	3	1.000
Overall	18	1.000

Table 2: Multi-document locality results. `locality_arxiv` (10 cases) is the primary metric; `detail` cases confirm anchor-paper retrieval is unaffected by the larger multi-paper pool.

Subset	N	R@1	R@5	MRR
All	18	0.167	0.444	0.269
Locality (all)	15	0.000	0.333	0.122
Locality (arXiv)	10	0.000	0.500	0.183
Detail (anchor)	3	1.000	1.000	1.000

fallback. Table 1 reports retrieval by reasoning type. We deliberately keep this benchmark small and frame it as a pilot: it validates the routing mechanism and gives a first read on answer quality, not that visual QA is solved.

5.4 Multi-Document Locality Benchmark

We construct an 18-case multi-document benchmark in the style of M3SciQA (Li et al. 2024): 15 *locality* cases requiring the system to identify which cited reference (not the anchor paper) contains the answer, and 3 *detail* cases answerable from the anchor alone as a sanity check. Of the locality cases, 10 reference arXiv-resolvable works (`locality_arxiv`, our primary metric); the remaining 5 reference pre-arXiv works or datasets (e.g., WMT, Dropout) that cannot currently be downloaded and always score zero.

The correct cited paper is recoverable within the top 5 results only half the time (50%) and essentially never ranks first (0%): with dozens of papers’ chunks merged into a single flat retrieval pool, a niche secondary paper must win on lexical match alone against the anchor paper’s typically larger, more topically central chunk set. Section 6 discusses this finding relative to M3SciQA’s comparable observation that even large foundation models struggle at this exact task.

A chunk-ID collision bug inflated an earlier version of this result. We initially measured Locality (arXiv) at R@5 0.800 / MRR 0.356. Per-paper chunk numbering restarts at `chunk_001` for every paper, and the retrieval scorer keyed its BM25 score cache by this non-globally-unique ID rather than list position, so merged multi-document chunks with colliding IDs silently inherited each other’s scores (verified: 15 of 34 chunks collided when merging two representative papers). The numbers above and in Table 3 are corrected after the fix, reported the same way we reported the tokenizer-fix regression in Section 6: as a genuine finding, not a number to quietly revise.

Table 3: Multi-document locality: ScholAR vs. a random-guess floor and an oracle upper bound (10 `locality_arxiv` cases).

System	R@1	R@5	MRR
Random floor (no signal)	0.125	0.625	0.340
ScholAR (flat retrieval)	0.000	0.500	0.183
Oracle (correct paper isolated)	0.000	0.700	0.289

Table 4: Visual grounding under two local models (18 cases each). Answer score is a keyword-overlap heuristic, not an LLM-judge score.

Model	Retrieval R@5	Answer Score	Fallback
<code>gemma4:12b</code>	1.000	0.731	0/18
<code>qwen3.5:9b</code>	1.000	0.812	0/18

To give this number honest context rather than leaving it unexplained, we compute two reference bounds on the 10 `locality_arxiv` cases (average 8.0 candidate secondary papers per case): a *random floor* — the expected recall of picking a candidate paper uniformly at random with no retrieval signal at all — and an *oracle* upper bound, where retrieval is restricted to only the anchor paper plus the single correct secondary paper (i.e., the paper-selection problem is assumed solved, isolating chunk-level retrieval precision *within* the correct paper). Table 3 reports all three.

Both bounds are, after the chunk-ID fix above, less favorable than we initially found: ScholAR’s R@5 (0.500) sits *below* the random-guess floor (0.625, average 8.0 candidate papers per case), and the oracle bound (correct paper isolated) still only reaches R@1 0.000 / MRR 0.289 — meaning even with paper-selection assumed solved, chunk-level retrieval still fails to rank the right chunk first in any case. Section 6 discusses what this revises about our earlier interpretation.

5.5 Local Model Comparison

Retrieval and faithfulness scoring in ScholAR are model-independent (BM25 and a fixed local sentence encoder, no LLM call in either path), so the only benchmark where local-model choice is measurable is visual grounding, where the vision model both reads the figure and writes the answer. We run the 18-case visual grounding benchmark (Section 5.3) under `gemma4:12b` (Gemma Team 2024) and `qwen3.5:9b` (Qwen Team 2023), holding everything else fixed. Table 4 reports retrieval grounding and a keyword-overlap answer-quality proxy. Both retrieve the correct figure in all 18 cases with no caption-only fallback, but the smaller `qwen3.5:9b` scores higher on answer quality than the larger `gemma4:12b`. We do not read this as smaller models being generally better — the benchmark is small and the proxy is lexical, not an LLM-judge — but it shows going fully local does not require the largest available model.

5.6 Does Reading the Image Help? A Caption-Only Ablation

Rather than implement a full ColPali-style (Faysse et al. 2024) visual embedding baseline, we run a cheaper, more direct ablation: send the identical prompt to the same local model *without* the figure image attached, scored with the same heuristic. Under `gemma4:12b`, caption-only scores **0.752** versus **0.731** for full vision — removing the image does not lower this particular score.

We do not read this as evidence that visual reading is unnecessary. Many questions ask about a value also restated in the paper’s prose near the figure, so a text-only answer can be correct without the image; and our proxy rewards lexical overlap with the *question* rather than grounded correctness, so a model reporting fine-grained visual details absent from the question wording is not credited for them. Separating these effects properly needs benchmark cases whose answer is genuinely absent from the prose plus a correctness-based judge, not a lexical-overlap proxy — we flag this as follow-up work rather than overstate what this ablation shows.

5.7 Baseline Systems

For retrieval, we compare five configurations: (1) **Keyword Overlap**: Jaccard-style term overlap with page-hint boost; (2) **BM25-Only**: standard BM25 with page-hint boost; (3) **BM25+Rerank (No PH)**: BM25 without page hints; (4) **BM25+Rerank (PH)**: full system with page hints (default); (5) **Dense-Only**: single-pass `all-MiniLM-L6-v2` cosine similarity, no BM25, no rerank, no page hints — the naive-RAG point in the lexical-to-dense taxonomy (Thakur et al. 2021). For faithfulness, we compare: (6) **BM25-primary**: the ScholAR production retriever; (7) **Hybrid BM25+Dense+RRF**: dense retrieval fused via Reciprocal Rank Fusion (Cormack, Clarke, and Buettcher 2009).

5.8 Retrieval Results

Table 5 shows the retrieval results. BM25-primary outperforms keyword overlap on R@1, R@3, and MRR, and ties it on R@5. Page hints do not change the aggregate MRR; the two page-hint-specific cases individually retrieve byte-identical rankings with and without hints, so we cannot claim a measurable benefit even per-case here. More broadly, none of Section 4.3’s reranking heuristics move BM25-Only, BM25+Rerank (No PH), and BM25+Rerank (PH) apart on any metric — they are identical to three decimal places. We report this plainly: the measured gain over the keyword baseline is attributable to BM25 itself, not to ScholAR’s own reranking layer, whose benefit this small, homogeneous benchmark cannot detect either way. Dense-Only unexpectedly beats every lexical baseline (R@1 0.786, R@3/R@5 1.000, MRR 0.881), contrary to BEIR’s finding that zero-shot dense retrieval trails BM25 (Thakur et al. 2021), likely an artifact of this benchmark’s small size and paraphrased queries. At faithfulness scale (51 cases, Table 6) hybrid, not dense-only, wins, so we do not read this as grounds to change the production retriever.

Table 5: Retrieval results on 14 manually annotated cases. PH = page hints. Best values in **bold**.

System	R@1	R@3	R@5	MRR
Keyword Overlap	0.571	0.786	0.929	0.687
BM25 Only	0.714	0.857	0.929	0.788
BM25+Rerank (No PH)	0.714	0.857	0.929	0.788
BM25+Rerank (PH)	0.714	0.857	0.929	0.788
Dense-Only	0.786	1.000	1.000	0.881

Table 6: SummaC-ZS faithfulness results on 51 oracle-claim cases. T1 = SummaC-ZS entailment fraction; SCR = whole-claim cosine similarity; KFP = Key Fact Presence; SCHR@K = Supporting Chunk Hit Rate.

Metric	BM25-primary	Hybrid+RRF
SummaC-ZS / T1 (entailment)	0.941	0.961
SCR / T2 (cosine similarity)	0.484	0.509
KFP / T3 (key fact recall)	0.966	0.979
Combined CFS	0.809	0.829
SCHR@1	0.373	0.392
SCHR@3	0.725	0.686
SCHR@5	0.824	0.922
Faithful (≥ 0.55 CFS)	48 / 51	49 / 51

5.9 Faithfulness Results

Table 6 reports SummaC-ZS faithfulness results on the 51-case benchmark. Hybrid BM25+Dense+RRF reaches Combined CFS of **0.829**, versus 0.809 for BM25-primary. More notably, hybrid raises SCHR@5 from 0.824 to **0.922** (+9.8 pp), confirming dense retrieval recovers chunks BM25 lexical matching misses. BM25-primary classifies 48 of 51 cases (94.1%) and Hybrid **49 of 51** (96.1%) as FAITHFUL (≥ 0.55 CFS). Both systems’ Tier 1 entailment (0.941 / 0.961) and KFP (0.966 / 0.979) are high, with SCR the primary differentiator between them.

5.10 Ablation: CFS vs. Retrieval Depth

Table 7 shows Combined CFS growing monotonically with more retrieved chunks, confirming evidence for complex claims is distributed across pages and Top-5 is necessary for full coverage. The hybrid system leads at every depth except a near-tie at Top-3 (BM25 0.799 vs. Hybrid 0.796, a 0.003 gap), with the largest hybrid lead at Top-4 (BM25 0.791 vs. Hybrid 0.817, a 0.026 gap).

5.11 Ablation: CFS by Claim Type

Table 8 breaks down Combined CFS by claim type under BM25-primary. Environmental and human evaluation claims score highest (0.918, 0.882), but with only 1 and 2 cases respectively, N is too small to draw a mechanism-level conclusion about why. Training-detail claims (0.668) are penalised by multi-hop evidence spread across sections; the two chronic BM25 failures (`attn_training_steps`: 0.242 UN-FAITHFUL; `llama_training_tokens`: 0.356 PAR-

Table 7: Combined CFS at each retrieval depth ($N = 51$ cases). Both improve monotonically; hybrid leads at every depth except a near-tie at Top-3.

Chunks	N	BM25	Hybrid
Top-1	51	0.658	0.669
Top-2	51	0.757	0.776
Top-3	51	0.799	0.796
Top-4	51	0.791	0.817
Top-5	51	0.809	0.829

Table 8: CFS breakdown by claim type, BM25-primary system, $N=51$.

Claim Type	N	BM25 CFS	Hybrid CFS
Environmental claim	1	0.918	0.918
Human evaluation	2	0.882	0.882
Result number	13	0.811	0.860
Conceptual claim	5	0.847	0.854
Technical claim	11	0.846	0.856
Architecture detail	10	0.830	0.845
Formula	1	0.876	0.876
Training detail	8	0.668	0.674

TIAL) both involve numerical claims spread across non-contiguous pages. A third failure, `rag_trivia_score` (0.235 UNFAITHFUL), is retrieval-induced rather than distributed-evidence: BM25 misses the supporting chunk entirely, while Hybrid recovers it (0.802 FAITHFUL) via the dense component — see the retrieval-precision/compound-term-matching tradeoff discussed below.

6 Discussion

Hybrid retrieval improves semantic recall. BM25-primary achieves Recall@5 of 0.929 and MRR of 0.788, confirming lexical matching over page-preserving chunks as a reliable foundation. Hybrid dense retrieval adds meaningful value on top: `SCHR@5` improves from 0.824 to 0.922 (+9.8 pp) and Combined CFS rises from 0.809 to 0.829 (+2.0 pp), driven by the dense component surfacing relevant chunks when exact BM25 terms are absent.

SummaC-ZS detects paraphrased entailment that ROUGE cannot. Our Tier 1 SummaC-ZS score is 0.941 for BM25-primary and 0.961 for Hybrid — higher than a purely lexical entailment check would likely yield on the same cases, for the reason given in Section 2.7 (sentence-level embedding similarity captures paraphrase, ROUGE does not). The SCR tier (whole-claim cosine, 0.484 BM25 vs. 0.509 Hybrid) is the primary differentiating signal between the two retrieval systems.

Training-detail failures and retrieval depth. The two non-FAITHFUL cases (Section 5.2) both involve numerical evidence spread across non-contiguous pages, validating structured extraction as highest-priority future work; the monotonic Top-1-to-Top-5 CFS gains (Table 7) confirm the $K=5$ default is well-motivated.

Tokenizer fixes trade off single-document precision against compound-term matching; hybrid retrieval absorbs the cost better than BM25 alone. We fixed a tokenizer bug uncovered during multi-document testing: token boundaries only start on a letter, so a fused compound proper noun (e.g. a figure or paper name written as one word) and its space-separated form reduce to different token sets, breaking exact-name retrieval. The fix is necessary for Sections 4.5 and 4.6, but perturbs BM25’s length normalisation globally: it cost BM25-only Recall@5 seven points ($1.00 \rightarrow 0.929$) and introduced one new faithfulness miss (`rag_trivia_score`). Hybrid BM25+Dense+RRF was materially more robust — barely moved, and still recovers the case BM25 now misses.

The multi-document pipeline is weak, and not only for the reason we first thought. The multi-document locality benchmark (Section 5.4) shows Recall@5 of 0.50 and Recall@1 of 0.00, both at or below the random-guessing floor over the same pool (Table 3). We had attributed this entirely to flat retrieval — a niche cited paper competing against the anchor paper’s larger, more central chunk set in one shared BM25 pool — and a strong oracle bound seemed to confirm that isolating the correct paper would mostly fix it. That oracle number was inflated by the chunk-ID collision bug (Section 5.4): corrected, it shows chunk-level retrieval failing to rank the right chunk first even with paper-selection solved ($R@1$ 0.00, MRR 0.289). So this is two compounding weaknesses, not one: cross-paper competition *and* a within-paper retrieval difficulty on these specific cross-document questions — mirroring M3SciQA (Li et al. 2024)’s finding that even large foundation models underperform humans here, though our corrected numbers suggest the difficulty runs deeper than paper selection alone. We still view per-paper-aware retrieval/reranking and the Recursive Language Model pattern (Zhang, Kraska, and Khattab 2025) as promising, but no longer sufficient on their own without diagnosing why oracle-setting retrieval underperforms.

Baseline scope. Section 2.2 explains why OpenScholar, PaperQA2 (Skarlinski et al. 2024), and SciRAG are discussed qualitatively, not head-to-head: none targets a consumer laptop, so a fair comparison needs a resource-matched setup, left to future work.

7 Future Work

The training-detail failure mode (Section 5.2) motivates structured extraction next: integrating Nougat (Blecher et al. 2023) for LaTeX-preserving extraction, evaluated against ColPali (Faysse et al. 2024)-style visual retrieval as an alternative to our caption-and-crop grounding. We also plan to evaluate the Recursive Language Model pattern (Zhang, Kraska, and Khattab 2025) for multi-document citation-chasing, our clearest open problem.

8 Conclusion

Scholar grounds scientific QA in page-preserving chunks and indirect citation IDs, extended to visual and multi-document QA. Retrieval, faithfulness, and visual grounding are strong; multi-document locality remains open.

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